

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460959.8871 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.047
Transit depth (Rp/Rs)^2	2.09 +/- 1.37 [%]
Orbital Inclination (inc)	83.11 +/- 2.0
Airmass coefficient 1 (a1)	0.77 +/- 0.28
Airmass coefficient 2 (a2)	0.23 +/- 1.03
Scatter in the residuals of the lightcurve fit is	36.96 %
Best Comparison Star	None
Optimal Aperture	4.18
Optimal Annulus	10.45
Transit Duration (day)	0.022 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.047 vs 0.1241 (deviation 0.34σ)
Duration fit vs published	0.022 d vs 0.1317 d (deviation 4.38σ)
Mid-time O–C	-8.7 min
Depth SNR	2.3
Residual scatter	36.96 %

Light curve

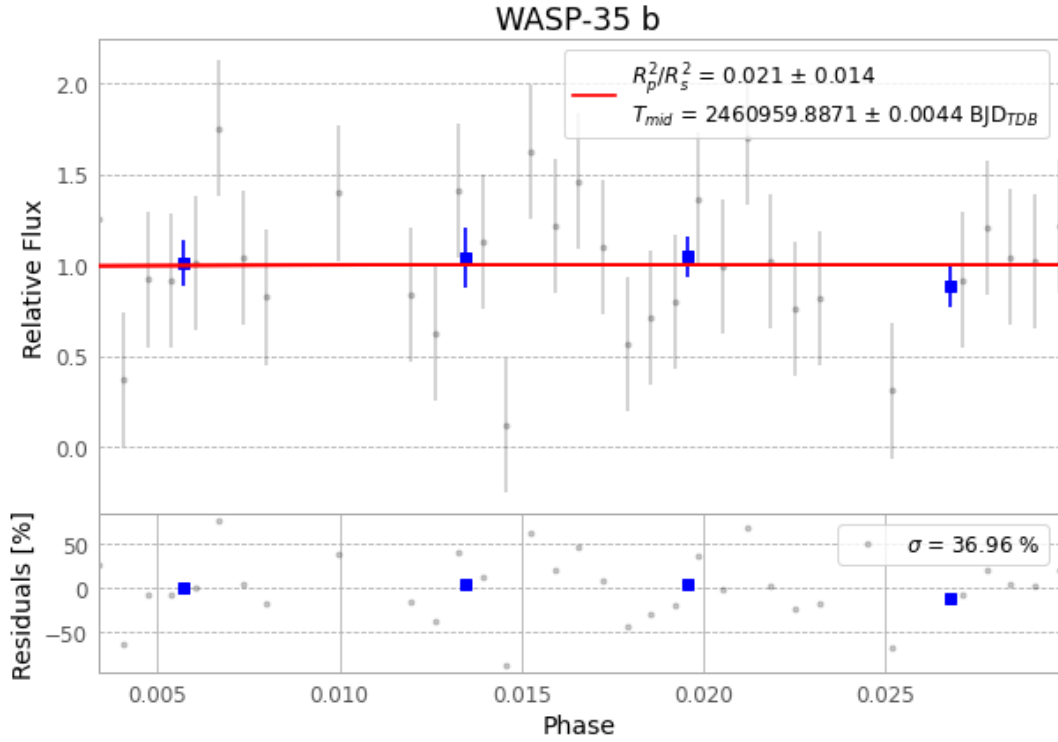


Figure 1 — FinalLightCurve_WASP-35 b_2025-10-11.png (SHA-256 52e635ba12c70723...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [572, 341], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 304ee9faf2dd4817...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

34 FITS frames (22 MB), WASP-35251011092716.FITS → WASP-35251011112716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-251011.log (57b0969e85f1...), FinalLightCurve_WASP-35 b_2025-10-11.png (52e635ba12c7...), FinalLightCurve_WASP-35 b_2025-10-11.csv (32d366fd1689...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:51Z.